<table>
<tr><td>1</td><td>
<table>
<tr><th>a</th><th>b</th><th>c</th><th>(abc)</th></tr>
<tr><td>6</td><td>1</td><td>3</td><td>(18)</td></tr>
<tr><td>4</td><td>2</td><td>4</td><td>(32)</td></tr>
<tr><td>2</td><td>3</td><td>5</td><td>(30)</td></tr>
</table>
Any one correct row for $b = 1$, $b = 2$ or $b = 3$.
Products do not need to be found for this mark.
</td><td>B1</td></tr>
<tr><td></td><td>Attempts the product abc for at least 2 valid combinations.</td><td>M1</td></tr>
<tr><td></td><td>Finds all three valid combinations with correct products seen and somewhere/shows why this is exhaustive and concludes. *</td><td>A1*</td></tr>
<tr><td></td><td></td><td>(3 marks)</td></tr>
</table>

2(a)	$f(\frac{5}{4}) = (2 - k \times \frac{5}{4})^5 = \frac{243}{32} \Rightarrow (2 - k \times \frac{5}{4}) = \sqrt[5]{\frac{243}{32}} \Rightarrow k = ...$	M1
	$\frac{3}{2} \Rightarrow \frac{5k}{4} = \frac{1}{2} \Rightarrow k = \frac{2}{5}$ *	A1*
		(2)
(b)	$\pm {}^5C_1 \times 2^4 \times \left(\pm \frac{2}{5}x\right)$ or $\pm {}^5C_2 \times 2^3 \times \left(\pm \frac{2}{5}x\right)^2$	M1
	$32 - 32x + \frac{64}{5}x^2$	A1A1
		(3)
(c)	$f'(x) = -32 + \frac{128}{5}x + ... \Rightarrow f'(0) = ...$	M1
	$f'(0) = -32$	A1ft
		(2)
		(7 marks)

3(a)(i)	$a_1 = \frac{1}{4}$	B1
(ii)	$a_2 = \frac{1}{4}$	B1
(iii)	$a_3 = 1$	B1
		(3)
(b)	$\frac{50}{2}[2+49]$ $(=1275)$ oe	M1A1
	$\sum_{n=1}^{50} \cos^2\left(\frac{n\pi}{3}\right) = 34 \times \frac{1}{4} + 16 \times 1$	M1
	$1275 + \frac{49}{2} = \frac{2599}{2}$	A1
		(4)
		(7 marks)

4(a)	$10 = \log_a 8 - \log_a 4 \Rightarrow \log_a 2 = 10$	M1
	$a^{10} = 2$	M1
	$a = 2^{1/10} = 1.07177...$ *	A1*
		(3)
(b)	$w = \log_{1.072}(t+5) - \log_{1.072} 4 \Rightarrow w = \log_{1.072}\left(\frac{t+5}{4}\right)$	M1
	$w = \log_{1.072}\left(\frac{t+5}{4}\right) \Rightarrow \frac{t+5}{4} = 1.072^w \Rightarrow t = ...$	M1
	$t = 4 \times 1.072^w - 5$	A1
		(3)
(b) ALT	$w = \log_{1.072}(t+5) - \log_{1.072} 4 \Rightarrow w + \log_{1.072} 4 = \log_{1.072}(t+5)$	M1
	$\Rightarrow t + 5 = 1.072^{w+\log_{1.072} 4}$	M1
	$\Rightarrow t = 1.072^{w+\log_{1.072} 4} - 5$	A1
(c)	$t = 4 \times 1.072^{15} - 5 = ...$	M1
	awrt 6.35	A1
		(2)
		(8 marks)

5(a)	$\cos\theta(3\cos\theta - \tan\theta) = 2 \Rightarrow \cos\theta(3\cos\theta - \frac{\sin\theta}{\cos\theta}) = 2$ or e.g. $\cos\theta(3\cos\theta - \tan\theta) = 2 \Rightarrow 3\cos^2\theta - \sin\theta = 2$ or e.g. $\cos\theta(3\cos\theta - \tan\theta) = 2 \Rightarrow 3\cos^2\theta - \frac{\sin\theta}{\cos\theta}\cos\theta = 2$	M1
	$3(1-\sin^2\theta) - \sin\theta = 2$	M1
	$3\sin^2\theta + \sin\theta - 1 = 0$ *	A1*
		(3)
(b)	$(\sin 2x =)\frac{-1 \pm \sqrt{13}}{6}$ (or awrt 0.43 and awrt –0.77 (or truncated –0.76))	M1A1
	$2x = \sin^{-1}(0.434...)$ or $2x = \sin^{-1}(-0.77...) \Rightarrow x = ...$	M1
	$-1.13, -0.438, 0.225, 1.35$	A1A1
		(5)
		(8 marks)

6(a)	$h = 0.5$	B1
	$\frac{1}{2} \times "0.5" \times \left[3 + 1.92 + 2(2.6833 + 2.4 + 2.1466)\right]$	M1
	4.845	A1
		(3)
(b)	$\int_0^2 2 - \frac{1}{4}x^2 \, \mathrm{d}x = \left[2x - \frac{x^3}{12}\right]_0^2 = \frac{10}{3}$	M1A1
	$\%\text{ of logo shaded } = \frac{"4.845" - "\frac{10}{3}"}{6}$	dM1
	$= 25.2(\%)$	A1
		(4)
		(7 marks)

7(a)	$\dfrac{12x^3(x-7)+14x(13x-15)}{21\sqrt{x}} = \dfrac{12x^4-84x^3+182x^2-210x}{21\sqrt{x}}$	M1
	$\dfrac{4}{7}x^{\frac{7}{2}}, -4x^{\frac{5}{2}}, +\dfrac{26}{3}x^{\frac{3}{2}}, -10x^{\frac{1}{2}}$	A1
	$\left(y=\right)\dfrac{4}{7}x^{\frac{7}{2}}-4x^{\frac{5}{2}}+\dfrac{26}{3}x^{\frac{3}{2}}-10x^{\frac{1}{2}}$	A1
		(3)
(b)	$y=\dfrac{4}{7}x^{\frac{7}{2}}-4x^{\frac{5}{2}}+\dfrac{26}{3}x^{\frac{3}{2}}-10x^{\frac{1}{2}}$ $\left(\dfrac{\mathrm{d}y}{\mathrm{d}x}=\right)2x^{\frac{5}{2}}-10x^{\frac{3}{2}}+13x^{\frac{1}{2}}-5x^{-\frac{1}{2}}$	M1A1ft
	$2x^3-10x^2+13x-5=0$ *	A1*
		(3)
(c)	$\left(2x^3-10x^2+13x-5\right)\div(x-1)=(2x^2\pm...x\pm...)$	M1
	$2x^2-8x+5$	A1
	$x=\dfrac{4\pm\sqrt{6}}{2}$	A1
		(3)
		(9 marks)

8(a)	$3a = \frac{a}{1-r} \Rightarrow r = ...$	M1
	$r = \frac{2}{3}$ *	A1*
		(2)
(b)	$ar - ar^3 = 16$	B1
	$\frac{10}{27}a = 16 \Rightarrow a = ...$	M1
	$a = 43.2$	A1
	$S_{10} = \frac{43.2(1-\left(\frac{2}{3}\right)^{10})}{1-\frac{2}{3}} = 127.4$	dM1A1
		(5)
		(7 marks)

9(a)	$y = x^3 - 5x^2 + 3x + 14 \Rightarrow \dfrac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 10x + 3 = 0$	M1
	Roots are $3, \dfrac{1}{3} \Rightarrow$ when $x = 3,\ y = "3"^3 - 5\times"3"^2 + 3\times"3" + 14 = ...$	dM1
	Centre is $(3, 5)$	A1
		(3)
(b)	At $A \quad y = 8$	B1
	$r^2 = (2-"3")^2 + ("8"-"5")^2 \quad (=10)$	M1
	$(x-3)^2 + (y-5)^2 = 10$	A1
		(3)
(c)	$\dfrac{"8"-"5"}{2-"3"} = ...(-3)$	M1
	$y - "8" = "\dfrac{1}{3}"(x-2)$	M1
	$y = \dfrac{1}{3}x + \dfrac{22}{3} \quad *$	A1*
		(3)
(d)	$\int_0^2 x^3 - 5x^2 + 3x + 14 \ \mathrm{d}x = ...\left(\dfrac{1}{4}x^4 - \dfrac{5}{3}x^3 + \dfrac{3}{2}x^2 + 14x\right)$	M1
	Area $= \left[\dfrac{1}{4}x^4 - \dfrac{5}{3}x^3 + \dfrac{3}{2}x^2 + 14x\right]_0^2 - \left(\dfrac{1}{2}\times\left(\dfrac{22}{3} + "8"\right)\times 2\right) = ...$ $\dfrac{1}{4}\times 16 - \dfrac{5}{3}\times 8 + \dfrac{3}{2}\times 4 + 14\times 2 - \dfrac{46}{3}$	dM1
	$\dfrac{74}{3} - \dfrac{46}{3} = \dfrac{28}{3}$	A1
		(3)
		(12 marks)

10(i)(a)	$2a$	B1
(b)	$\log_2\left(\frac{\sqrt{3}}{16}\right) = \log_2\sqrt{3} - \log_2 16, = \frac{1}{2}a - 4$	M1A1
		(3)

(ii)	Examples: $3^x \times 2^{x+4} = 6 \Rightarrow \log_2 3^x + \log_2 2^{x+4} = \log_2 6$ or $3^x \times 2^{x+4} = 6 \Rightarrow 3^x \times 2^x \times 2^4 = 6 \Rightarrow \log_2 3^x + \log_2 2^x + \log_2 2^4 = \log_2 6$ or $3^x \times 2^{x+4} = 6 \Rightarrow 3^x \times 2^{x+3} = 3 \Rightarrow \log_2 3^x + \log_2 2^{x+3} = \log_2 3$ or $3^x \times 2^{x+4} = 6 \Rightarrow 3^x \times 2^x = \frac{3}{8} \Rightarrow \log_2 3^x + \log_2 2^x = \log_2 \frac{3}{8}$ or $3^x \times 2^{x+4} = 6 \Rightarrow 3^{x-1} \times 2^{x+3} = 1 \Rightarrow \log_2 3^{x-1} + \log_2 2^{x+3} = \log_2 1$	M1
	Examples: $x\log_2 3 + (x+4)\log_2 2 = \log_2 6 \Rightarrow x(\log_2 3 + \log_2 2) = \log_2 6 - 4 \Rightarrow x = ...$ or $x\log_2 3 + x\log_2 2 + 4 = \log_2 6 \Rightarrow x(\log_2 3 + \log_2 2) = \log_2 6 - 4 \Rightarrow x = ...$ or $x\log_2 3 + (x+3)\log_2 2 = \log_2 3 \Rightarrow x(\log_2 3 + \log_2 2) = \log_2 3 - 3\log_2 2 \Rightarrow x = ...$ or $x\log_2 3 + x\log_2 2 = \log_2 \frac{3}{8} \Rightarrow x(\log_2 3 + \log_2 2) = \log_2 \frac{3}{8} \Rightarrow x = ...$ or $(x-1)\log_2 3 + (x+3)\log_2 2 = 0 \Rightarrow ax + x = a - 3 \Rightarrow x = ...$	dM1
	$\log_2 a^b = b\log_2 a$	B1
	$x = \frac{a-3}{a+1}$	A1
		(4)
		(7 marks)